

Soft Computing Assignment 2 – Traffic FIS Report

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Date: 20/09/2025

1. Introduction

Traffic congestion is a common urban challenge, and effective traffic light control can help reduce delays at intersections. This assignment applies **Fuzzy Inference Systems (FIS)** to model traffic light timing using real-world considerations.

Two fuzzy systems were developed:

- **FIS1 (Balanced Priority):** Designed to minimize the difference in waiting times between the main street and the side street.
- **FIS2 (Main Priority):** Designed to minimize waiting time for the main street, even if this increases waiting time for the side street.

Both systems were implemented in Python using `scikit-fuzzy` and tested on a pre-compiled traffic simulator.

2. FIS1 – Balanced Priority

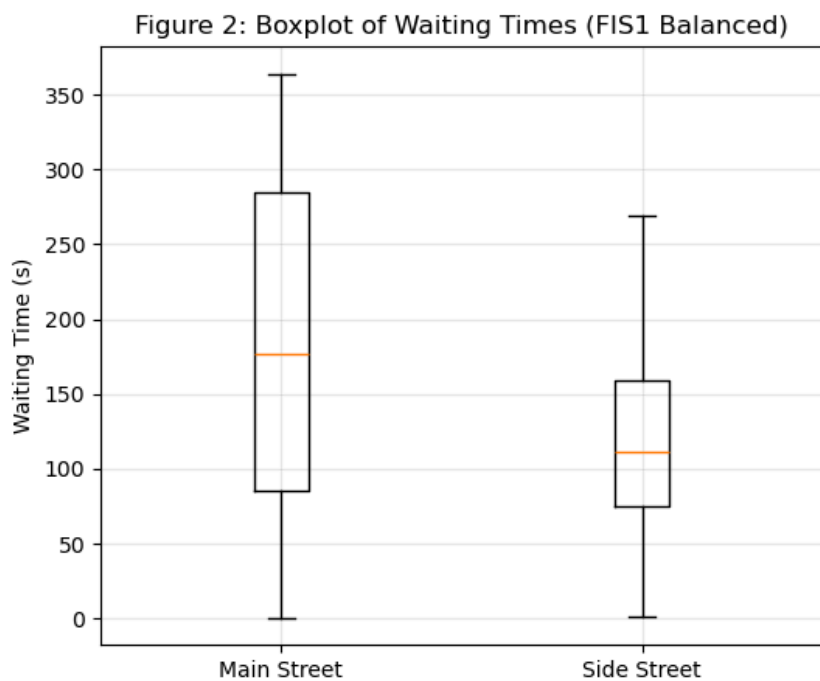
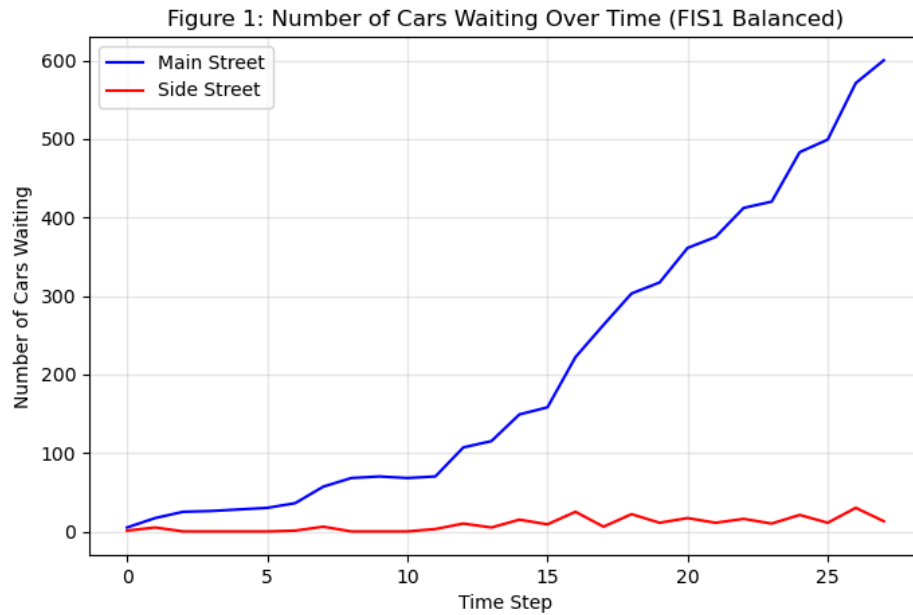
Key Aspects

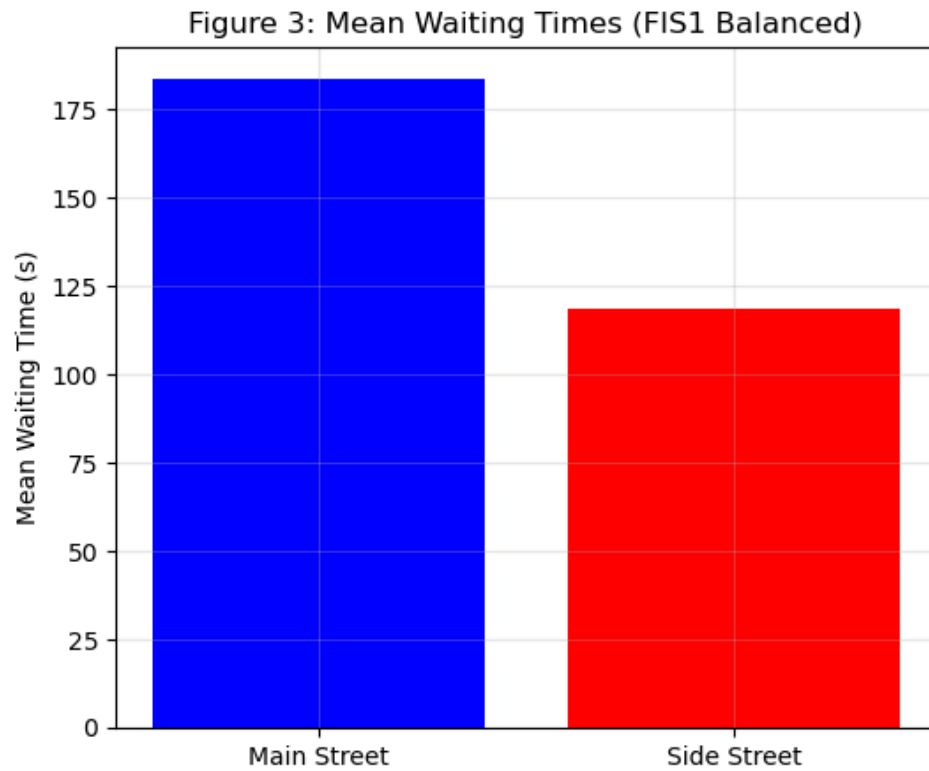
- **Inputs:**
 - Incoming traffic (main street).
 - Waiting traffic (side street).
- **Outputs:**
 - Main street green time.
 - Side street green time.
- **Membership Functions:** Designed with moderate overlap. This ensures that as one road's demand grows, the other road still receives a fair allocation of green time.
- **Rule Base:** Structured to balance waiting times rather than heavily biasing one road.

Results

- **Mean waiting times (from simulator):**
 - Main street: ~211.7 seconds

- Side street: ~141.4 seconds
- Ratio (main/side): ~1.5
- **Figures:**





- **Test Cases (Table 1):**

Incoming (main)	Waiting (side)	Main Green (s)	Side Green (s)	Ratio
10	5	20	5.4	3.69
50	30	60	21.9	2.73
80	70	99.6	21.7	4.6
0	90	60	40	1.5

90	10	99.6	5.4	18.38
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Comments

- The system narrows the gap between waiting times compared to untuned rules.
- However, main street sometimes waits longer than the side street.
- Goal achieved: **balance between the two streets**.

3. FIS2 – Main Street Priority

Key Aspects

- **Inputs & Outputs:** Same as FIS1.
- **Membership Functions:** Tuned to extend main street greens, especially at medium or high traffic levels.
- **Rule Base:** Explicitly biases main street by giving it “long” or “very long” greens in most scenarios. Side street only receives longer greens if main street is low and side street is highly congested.

Results

- **Mean waiting times (from simulator):**
 - Main street: ~208.4 seconds
 - Side street: ~258.3 seconds
 - Ratio (main/side): ~0.81
- **Figures:**

Figure 4: Number of Cars Waiting Over Time (FIS2 Aggressive)

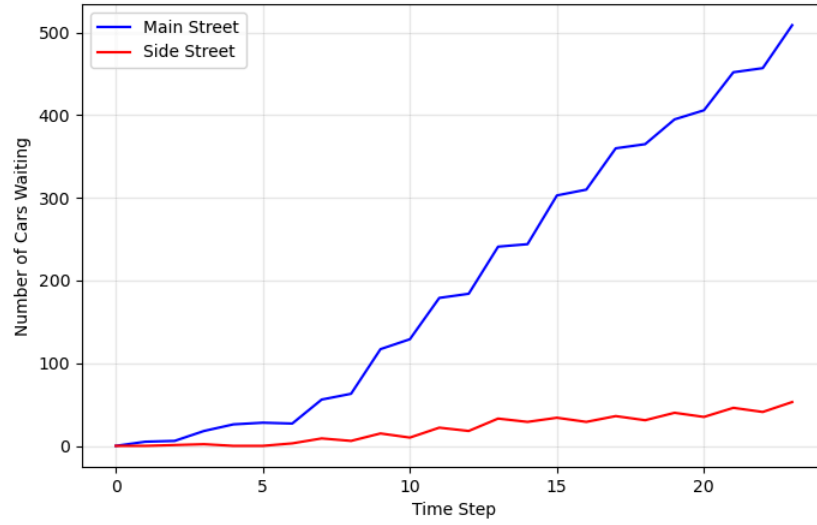


Figure 5: Distribution of Waiting Times (FIS2 Aggressive)

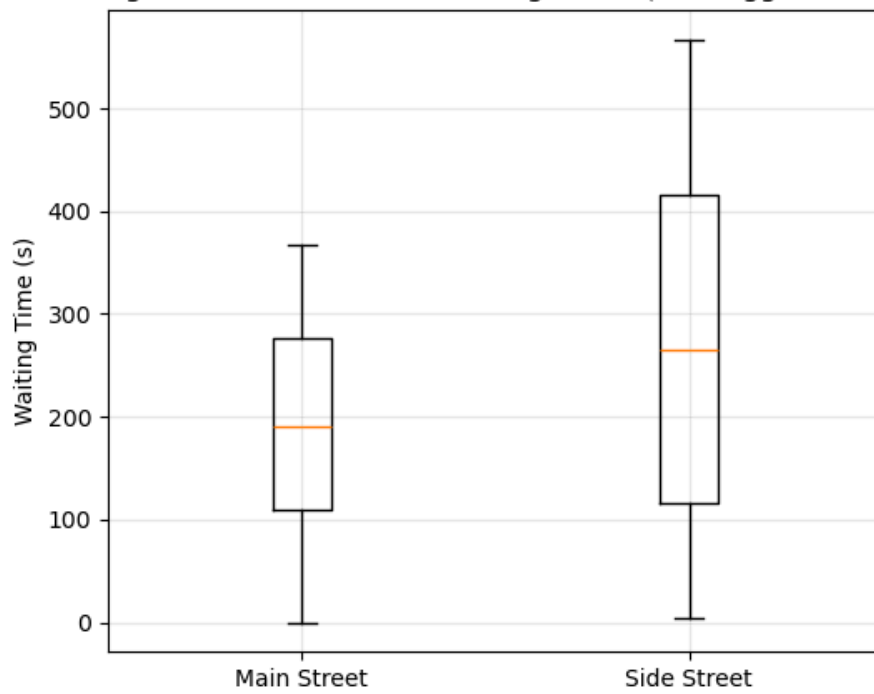
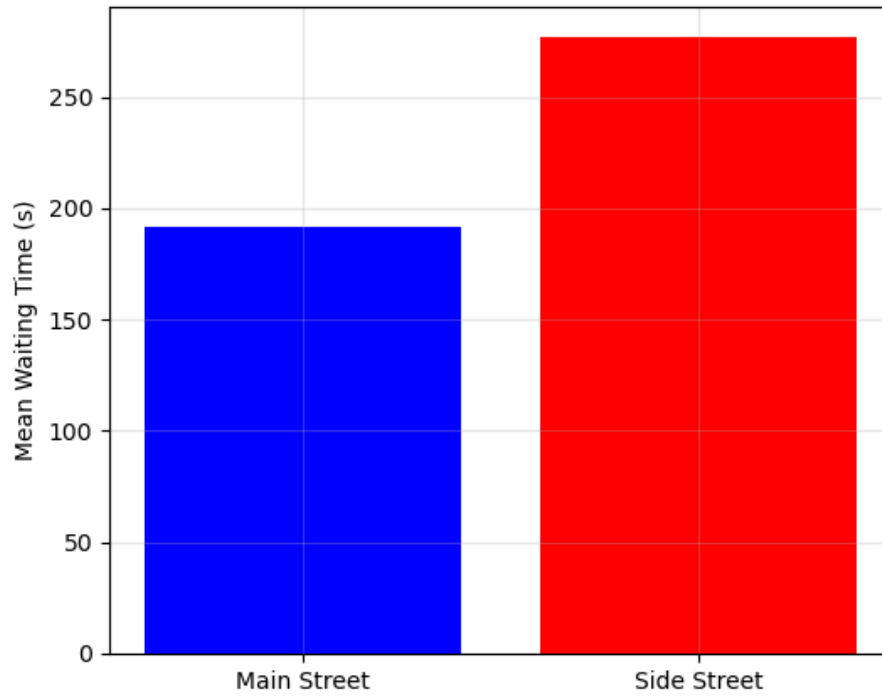


Figure 6: Mean Waiting Times (FIS2 Aggressive)



- **Test Cases (Table 2):**

Incoming (main)	Waiting (side)	Main Green (s)	Side Green (s)	Ratio
10	5	16.7	5.4	3.08
50	30	100	20	5
80	70	112.2	5.4	20.72
0	90	60	35	1.71

90	10	112.2	5.4	20.72
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Comments

- Main street wait time is consistently lower than side street wait time.
- The trade-off is that the side street experiences very long delays.
- Goal achieved: **main street is prioritized**.

4. Comparison of FIS1 vs FIS2

- **Observations:**
 - FIS1: More balanced waiting times.
 - FIS2: Main street has clear priority, but at the expense of fairness.
- Which system is “better” depends on policy:
 - Use **FIS1** if fairness is required.
 - Use **FIS2** if main street efficiency is the main priority.

5. Conclusion

This study demonstrated the use of fuzzy logic in traffic light timing. Two systems were implemented with distinct goals:

- **FIS1 (Balanced Priority):** Achieved fairer distribution of waiting times between streets.
- **FIS2 (Main Priority):** Achieved shorter waiting times for the main street, but side street performance worsened.

Future improvements:

- Hybrid system that dynamically switches between balance and priority depending on time of day.
- Adaptive learning to fine-tune membership functions based on live traffic data.